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**Key Points:**

- Local finite-amplitude wave activity (LWA) is an objective diagnostic of midlatitude weather events
- The LWA climatology and its variability associated with the AO agree with the blocking frequency
- Robust trends in wave activity can be identified at regional scale, but not at hemispheric scale

**Supporting Information:**

- Supporting Information S1

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# Local finite-amplitude wave activity as an objective diagnostic of midlatitude extreme weather

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**Abstract** Midlatitude extreme weather events are responsible for a large part of climate-related damage, yet our understanding of these extreme events is limited, partly due to the lack of a theoretical basis for midlatitude extreme weather. In this letter, the local finite-amplitude wave activity (LWA) of Huang and Nakamura (2015) is introduced as a diagnostic of the 500 hPa geopotential height for characterizing midlatitude weather events. It is found that the LWA climatology and its variability associated with the Arctic Oscillation agree broadly with the previously reported blocking frequency in the literature. There is a strong seasonal and spatial dependence in the trends of LWA in recent decades. While there is no observational evidence for a hemispheric-scale increase in wave amplitude, robust trends in wave activity can be identified at the regional scales, with important implications for regional climate change.

## 1. Introduction

Midlatitude extreme weather events are responsible for a large part of climate-related damage [*Hegerl et al.*, 2007], yet our understanding of these extreme events is limited, partly due to the lack of a theoretical basis for midlatitude extreme weather. While high-impact midlatitude weather events (e.g., blocking, heat waves, or atmospheric rivers) are generally manifested in the form of large-amplitude Rossby waves, the interpretations are often restricted to the linear Rossby wave dispersion relationship [e.g., *Coumou et al.*, 2015] that does not account for the effect of wave amplitude saturation on Rossby wave breaking (RWB) and dissipation (see *McIntyre and Palmer* [1983] and also discussion about equation (3) in section 2). In this letter, we introduce the local finite-amplitude wave activity (LWA) of *Huang and Nakamura* [2015] as a diagnostic of midlatitude weather events.

Studies to date have used the pattern of atmospheric circulation to characterize midlatitude weather extremes. Blocking has been identified by the meridional reversal of the 500 hPa geopotential height ($Z_{500}$) gradient [e.g., *Tibaldi and Molteni*, 1990; *Masato et al.*, 2013] or the tropopause potential temperature gradient [e.g., *Pelly and Hoskins*, 2003; *Woollings et al.*, 2008]. Atmospheric rivers are related to the local reversal of the upper tropospheric potential vorticity (PV) gradient [e.g., *Ryoo et al.*, 2013; *Payne and Magnusdottir*, 2014]. Midlatitude jet stream meandering and associated cold air outbreaks are measured by the meridional displacement of $Z_{500}$ [e.g., *Francis and Vavrus*, 2012], and it is important to separate the change in wave amplitude from the shift in $Z_{500}$ due to global warming [*Barnes*, 2013]. In spite of the usefulness of these metrics in identifying extreme events, it remains unclear how to quantify them with respect to large-scale atmospheric dynamics.

Recent advances in the theory of finite-amplitude wave activity (FAWA) [*Nakamura and Zhu*, 2010; *Nakamura and Solomon*, 2011; *Methven*, 2013; *Chen and Plumb*, 2014; *Lu et al.*, 2015] have linked the pattern of atmospheric circulation to large-scale wave dynamics mathematically through the divergence theorem. Defining wave activity by the areal displacement of a conservative quantity like PV, one can relate the change in wave activity to the eddy fluxes across the boundaries of the areal displacement. It has been shown that the FAWA provides a useful prediction for the bifurcation of anticyclonic and cyclonic RWB [*Solomon et al.*, 2012] and that the changes in the irreversible mixing of PV are critical to our understanding of the zonal mean circulation response to tropical heating and global warming-like climate forcing [*Chen et al.*, 2013; *Sun et al.*, 2013; *Lu et al.*, 2014]. Recently, *Huang and Nakamura* [2015] have generalized FAWA to LWA as a function of longitude, and LWA can capture the local wave amplification associated with the blocking episode during Hurricane Sandy. This development allows us to use LWA to further characterize weather extremes at the regional scales.

The paper is organized as follows. Section 2 reviews the basic theory of wave activity. Then, the LWA formalism is implemented in section 3 to characterize $Z_{500}$ during selected midlatitude extreme weather events in the reanalysis. In section 4, we will describe the LWA climatology, the pattern associated with the Arctic Oscillation (AO), and the linear trends in LWA in recent decades. Conclusions are provided in section 5.

## 2. Theory of Wave Activity

Consider a dynamical quantity, $q$, that is solely or at least predominantly advected by the two-dimensional nondivergent flow (i.e., $\nabla \cdot \mathbf{v} = 0$)

$$\frac{\partial q}{\partial t} + \mathbf{v} \cdot \nabla q = \dot{q}, \tag{1}$$

where $\mathbf{v} = (u, v)$ and $\nabla$ are the horizontal velocity and horizontal gradient operator, respectively. $\dot{q}$ denotes the processes other than the horizontal advection. The flow can be either wavy or turbulent. A snapshot of the $q$ distribution stirred by the flow is illustrated schematically in Figure 1. This resembles the wavy pattern for an isopleth of $Z_{500}$ (contour in Figure 2, $Z_{500}$ can be thought of as the $\ln p$-weighted average of temperature, and the corresponding eddy flux is the $\ln p$-weighted average of eddy heat flux (see equation (9))), upper tropospheric potential vorticity (PV) or atmospheric rivers during extreme weather events.

Defining the Eulerian wave activity as [e.g., *Vallis*, 2006]

$$A^E = \frac{\overline{q'^2}}{2}\left(\frac{1}{a}\frac{\partial \overline{q}}{\partial \phi}\right)^{-1}, \tag{2}$$

where overbars denote the zonal means, primes are the deviations therefrom, $a$ denotes Earth's radius, and $\phi$ is latitude, the wave activity is related to the eddy flux by the relationship (see equation (S5) in Text S1 in the supporting information)

$$\frac{\partial A^E}{\partial t} + \overline{v'q'} + \left(\frac{1}{a}\frac{\partial \overline{q}}{\partial \phi}\right)^{-1} \nabla \cdot \left(\frac{\overline{\mathbf{v}'q'^2}}{2}\right) = \left(\frac{1}{a}\frac{\partial \overline{q}}{\partial \phi}\right)^{-1} [\overline{q'(\dot{q})'}]. \tag{3}$$

In the small-amplitude limit, the cubic term on the left-hand side can be ignored. For pure horizontal advection (i.e., $\dot{q} = 0$), this leads to a relationship between the wave activity tendency and eddy flux. If $q$ is materially conserved, $q' = -\delta\phi\, \partial\overline{q}/\partial\phi$, where $\delta\phi$ is the meridional displacement in the $q$ contour. The wave activity can be rewritten as

$$A^E = \frac{a\overline{(\delta\phi)^2}}{2}\left(\frac{\partial \overline{q}}{\partial \phi}\right). \tag{4}$$

During extreme events, however, the cubic term in equation (3) can dominate the wave activity equation, leading to the homogenization of the $q$ field. This nonlinear effect is ignored in deriving the linear Rossby wave dispersion relationship, but it is accounted for by the finite-amplitude wave activity formalism described below.

For simplicity, we assume that $q$ approximately decreases with latitude like $Z_{500}$ or atmospheric moisture in the Northern Hemisphere (NH), while allowing overturning or cutoff of contours. For a given value of $q = Q$, the area bounded by the $Q$ contour toward the North Pole is $S(Q) = \int\int_{q \le Q} a^2 \cos\phi \mathrm{d}\lambda \mathrm{d}\phi$, where $\lambda$ is longitude. Then, we can define an equivalent latitude $\phi_e$ after an area-conserving adjustment of the $Q$ contour to a zonally symmetric state, establishing a monotonic relationship [*Butchart and Remsberg*, 1986]

$$\phi_e(Q) = \arcsin\left[1 - \frac{S(Q)}{2\pi a^2}\right]. \tag{5}$$

Introducing an eddy term as $\hat{q} \equiv q - Q$ and separating the southward and northward displacements in the $Q$ contour, the southern and northern LWA at the longitude $\lambda$ and latitude $\phi = \phi_e$ can be defined as (see Figure 1a)

$$A_S(\lambda, \phi_e) = \frac{a}{\cos\phi_e}\int_{\hat{q}\le 0,\ \phi\le\phi_e(Q),\ \lambda=\mathrm{const}} \hat{q}\cos\phi \mathrm{d}\phi$$
$$A_N(\lambda, \phi_e) = \frac{a}{\cos\phi_e}\int_{\hat{q}\ge 0,\ \phi\ge\phi_e(Q),\ \lambda=\mathrm{const}} \hat{q}\cos\phi \mathrm{d}\phi. \tag{6}$$

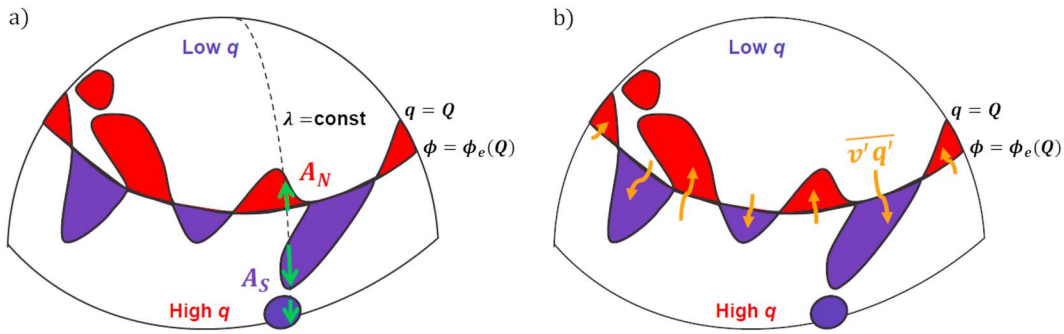


**Figure 1.** Schematics of a quasi-conservative dynamical quantity $q$ associated with idealized midlatitude extreme weather events. The blue shading denotes cyclonic wave activity to the south of the latitude $\phi_e$, $A_S$, and red shading describes anticyclonic wave activity to the north, $A_N$. Blue and red regions have equal areas such that the latitude $\phi_e$ is the equivalent latitude after an area-conserving adjustment of the $Q$ contour to a zonally symmetric state. (a) Green arrows indicate the paths along which the eddy term $\hat{q} \equiv q - Q$ is integrated for the LWA at $(\lambda, \phi_e)$. (b) Orange arrows depict the eddy fluxes producing the meridional areal displacement in the $Q$ contour.

Here northern/southern refers to the location relative to the equivalent latitude. Both $A_S$ and $A_N$ include the $\hat{q} = 0$ line to avoid any missing value for either $A_S$ or $A_N$ when $\hat{q} = 0$ at a longitude. For the typical midlatitude westerly jet in the NH, the northward excursion of the jet stream indicates an anomalous anticyclonic circulation, and the southward meandering corresponds to an anomalous cyclonic circulation. By construction, $A_S \leq 0$ in the NH, describing the cyclonic wave activity residing to the south of the equivalent latitude $\phi_e$, and $A_N \geq 0$, representing the anticyclonic wave activity to the north. In practice, for a latitude $\phi_e$, we first find the corresponding contour of $q = Q$ by inverting the monotonic relationship $\phi_e(Q)$ that is defined by equation (5). The line integral for the southern LWA, defined in equation (6), is computed at the longitude $\lambda$ by box-counting $\hat{q} = q - Q$ in the southern grid boxes relative to the latitude $\phi_e$ that satisfy $\hat{q} \leq 0$. Similarly, the northern LWA is calculated in the northern grid boxes satisfying $\hat{q} \geq 0$.

It follows that the total LWA can be defined as in *Huang and Nakamura* [2015]

$$A(\lambda, \phi_e) \equiv \mathcal{A}(q) = A_S - A_N, \tag{7}$$

where $\mathcal{A}(X)$ denotes an eddy operator for a given variable $X$. Hence, for the NH, $A(\lambda, \phi_e) \leq 0$, taking the difference between the negative $\hat{q}$ (blue in Figure 1) integrated over the southward flow and the integral of positive $\hat{q}$ (red) over the northward air flow, and therefore, it naturally measures the waviness of the dynamical field. Because the blue and red regions have equal areas, the zonally averaged LWA, $\overline{A(\lambda, \phi_e)}$, is identical to the FAWA [i.e., *Nakamura and Solomon*, 2011, equation (4)].

Since $Z_{500}$ or atmospheric moisture increases with latitude in the Southern Hemisphere (SH), the sign convention would have to be adjusted accordingly in the SH. In the conservative and small-amplitude limit, equation (4) suggests that the FAWA may be written as $\overline{A} = 0.5a(\partial Q/\partial \phi_e)\overline{(\delta\phi)^2}$, where $\delta\phi(\lambda)$ is the meridional displacement of the $Q$ contour. This shows the sign of LWA depends on that of $\partial Q/\partial \phi_e$. For a typical meridional height gradient $\partial Z_{500}/\partial \phi_e \sim 80\text{m}/10^\circ$, a meridional shift in $Z_{500}$ by $\delta\phi \sim 15^\circ$latitude implies a change in wave activity by $\Delta A \sim 10^8 \text{m}^2$.

Applying the eddy operator $\mathcal{A}(X)$ to equation (1), the FAWA is mathematically related to the lateral eddy fluxes across the latitude $\phi_e$ through the divergence theorem (see Figure 1b and the proof for $\overline{\mathcal{A}(\mathbf{v} \cdot \nabla q)} = \overline{v'q'}$ in supporting information Text S2) [*Nakamura and Zhu*, 2010; *Chen and Plumb*, 2014; *Lu et al.*, 2015]

$$\frac{\partial \overline{A}}{\partial t} + \overline{v'q'} = \overline{\mathcal{A}(\dot{q})}. \tag{8}$$

As such, the FAWA differs from an Eulerian quantity (e.g., $A^E$ in equation (2)) but is a Lagrangian measurement, and it links the meandering of atmospheric circulation to eddy fluxes irrespective of the nonlinearity of the flow. Compared with the Eulerian formula in equation (3), this is an exact balance between the zonally averaged wave activity and eddy flux for pure horizontal advection. While the eddy PV flux may be written as the divergence of wave activity flux in the linear limit, this cannot be generalized to other dynamical quantities. Furthermore, LWA gives the longitudinal distribution of FAWA, and the zonal advection of wave activity should be taken into account for the LWA budget as a function of longitude [see *Huang and Nakamura*, 2015].

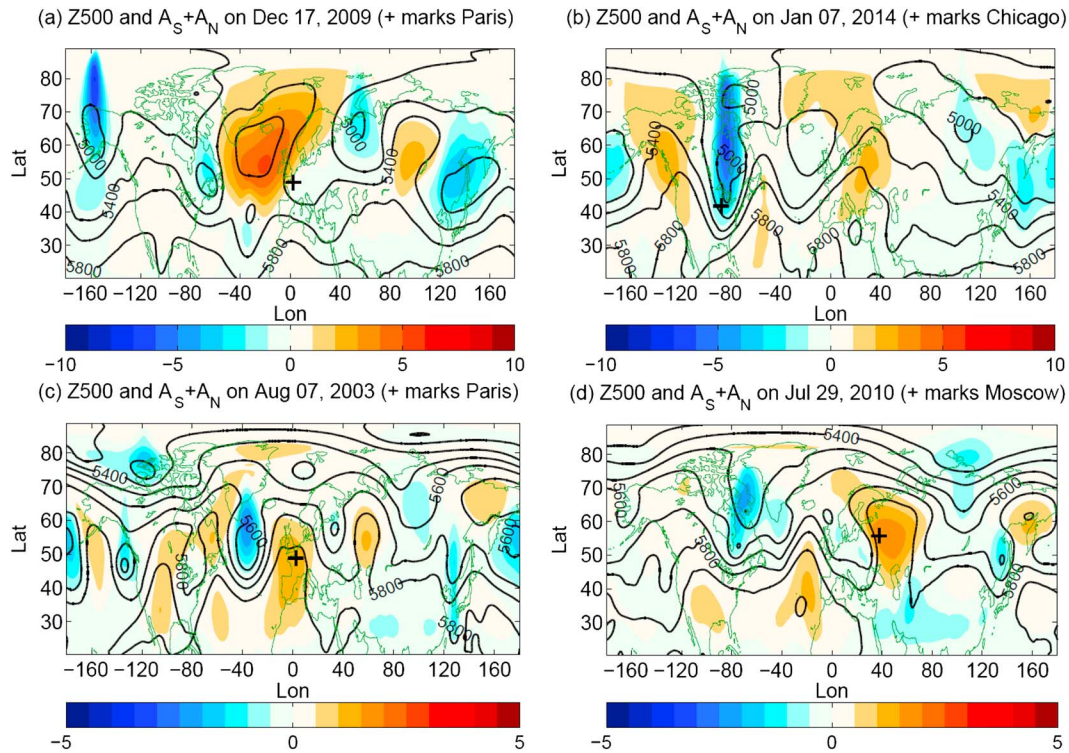


**Figure 2.** Geopotential height (500 hPa; contour) and corresponding southern and northern LWA (shading), $A_S + A_N$, for cold air outbreaks (a) in western Europe on 0000 UTC 17 December 2009 and (b) in North America on 0000 UTC 7 January 2014 and for summer heat waves (c) in western Europe on 0000 UTC 7 August 2003 and (d) in Russia on 0000 UTC 29 July 2010. Cities that were strongly influenced by individual events are marked by the bold plus sign. Unit: meters for height and $10^8$ m$^2$ for LWA. Notice that this differs from the total wave activity $A_S - A_N$ in equation (7) that emphasizes the wave amplitude, not the sign.

## 3. Application to Individual Midlatitude Weather Events

In this section, LWA is used to describe four midlatitude weather events. While the generalized EP flux [*Huang and Nakamura*, 2015] requires the conservation of PV and there is not such a conservation law of $Z_{500}$, *Masato et al.* [2013] found that the blocking frequency defined by $Z_{500}$ is consistent with the frequency detected using the upper tropospheric PV. In the observations, *Butchart et al.* [1989] found that $Z_{500}$ is approximately linearly related to the PV at 500 hPa, and thus, the former may be used as a proxy for the latter. As $Z_{500}$ is widely used to characterize synoptic weather events, the diagnostic is applied to $Z_{500}$ rather than the PV used in *Huang and Nakamura* [2015]. We employ the European Centre for Medium-Range Weather Forecasts's Era-Interim reanalysis (ERA-I) from January 1979 to December 2014 [*Dee et al.*, 2011]. The LWA analysis is performed for 4-times daily $Z_{500}$ from 20°N to the North Pole. The deep tropics are excluded from the analysis, because tropical $Z_{500}$ is strongly influenced by ageostrophic motions and the nondivergent assumption in section 2 is no longer valid (see *Chen and Plumb* [2014] for a possible generalization to ageostrophic flow). The robustness of the results from ERA-I is checked against the National Centers for Environmental Prediction-Department of Energy (NCEP-DOE) Reanalysis 2 (NCEP2) [*Kanamitsu et al.*, 2002] that is included in supporting information Figures S1 and S2.

The use of $Z_{500}$ for LWA is also amenable to an explicit relation to the meridional eddy heat flux. Heuristically, the temperature tendency in the lower troposphere during midlatitude large-scale extreme events is dominated by the horizontal temperature advection, $\partial T/\partial t = -\mathbf{v}_g \cdot \nabla T + \dot{T}$, where $\mathbf{v}_g$ denotes the geostrophic flow and $\dot{T}$ represents all other processes. Given the hypsometric equation $Z_{500} = (R/g)\int_{500\text{hPa}}^{SLP} T \mathrm{d}\ln p$, (see equation (1.22) in *Holton* [2004], and *SLP* is the sea level pressure), the theory in section 2 suggests that the tendency of wave activity in $Z_{500}$ is related to the eddy heat flux in the lower troposphere by

$$\frac{\partial \overline{A_{Z_{500}}}}{\partial t} \approx -\frac{R}{g}\int_{500\text{hPa}}^{SLP} \overline{v_g' T'} \mathrm{d}\ln p + \frac{R}{g}\left[\overline{\mathcal{A}\left(\int_{500\text{hPa}}^{SLP} \dot{T} \mathrm{d}\ln p\right)}\right]. \tag{9}$$

This indicates that the large wave activity in $Z_{500}$ is associated with an increased poleward heat flux, and thus, large $A_{Z_{500}}$ is expected to be closely related to regional temperature extremes. Since $Z_{500}$ is not a conservative quantity for adiabatic and frictionless flow, other processes such as the advection by the mean meridional circulation (MMC) can also alter the wave activity tendency. For example, the recent study by *Nie et al.* [2014] showed that both poleward eddy heat flux and RWB are important for the zonal wind variability in the SH, and the latter can induce an MMC that can impact the thermal structure of the atmosphere. Here we focus on using LWA as a diagnostic tool in this letter, leaving the comparison between the eddy heat flux and MMC for future investigation.

Figure 2 displays (contours) $Z_{500}$ and the corresponding (shading) northern and southern LWA for two cold air outbreak events in western Europe in mid-December 2009 and in North America in early January 2014 and for two heat wave events in western Europe in early August 2003 and in Russia in late July 2010. The definition in equation (6) indicates that $A_N \geq 0$, corresponding to a northward displacement of $Z_{500}$ and anticyclonic wave activity, and that $A_S \leq 0$, corresponding to a southward movement in height and cyclonic wave activity. Hence, $A_S + A_N$ gives a natural description of the meridional meandering of $Z_{500}$, with positive values corresponding to ridges and negative corresponding to troughs. (Note that because $A_S \leq 0$ and $A_N \geq 0$, there may be a cancelation between $A_S$ and $A_N$ if both of them are nonzero.)

In spite of the chaotic nature of the atmosphere, the LWA evidently highlights the high-impact weather events and their regional impacts, as in *Huang and Nakamura* [2015]. The blocking high over the Atlantic Ocean on 0000 UTC 17 December 2009, as indicated by the large northern LWA in Figure 2a, leads to the equatorward cold air advection largely responsible for the extremely cold temperature in Paris (marked by the plus symbol). In contrast, the synoptic low over the northwest Atlantic Ocean on 7 January 2014, depicted by the large southern LWA in Figure 2b, is linked to the cold temperature extreme in Chicago. Since many, if not all, blocking detection algorithms seek to identify either poleward blocking highs or persistent equatorward cutoff lows (see discussion in the appendix in *Masato et al.* [2013]), these features in the two above cases are well captured by northern and southern wave activities, making LWA a natural diagnostic of atmospheric blocking. Similarly, the blocking highs over Eurasia, indicated by the large northern LWA in Figures 2c and 2d, contribute substantially to the daily characteristics of heat wave events in Paris on 7 August 2003 and in Moscow on 29 July 2010. The impact of the large LWA is exacerbated by the long buildup and persistence of these events (not shown here) [e.g., *Fischer et al.*, 2007].

Since these extreme events are characterized by very large meridional displacements in the height contours (on the order of 30° latitude), the linear Rossby wave dispersion relationship is expected to break down for the propagation or decay of these events because of ignoring the large cubic term in equation (3). In contrast, by virtue of the divergence theorem in deriving equation (8) (see the proof of equation (S15) in supporting information Text S2), the LWA takes into account these nonlinear effects. Therefore, LWA outweighs linear theory as quantification of individual extreme weather events. Similarly, the events identified by the large LWA amplitude have a quantitative relation to the eddy flux by equation (8), whereas the large anomaly in $Z_{500}$ may have a complicated relation to the eddy flux due to the cubic term in equation (3). This could be important in explaining the change in wave amplitude in a changing climate, as the change in eddy energy flux can be well constrained by the global energy balance.

## 4. Wave Activity Climatology, Variability, and Trends

Having illustrated LWA for individual weather events, we proceed to describe the LWA climatology for $Z_{500}$, the pattern associated with the AO, and the linear trends in recent decades. As the LWA for $Z_{500}$ is negative in the NH, the inverted LWA is presented to indicate wave amplitude.

Figure 3 (left column) gives the 1979–2014 climatology of (shading) inverted LWA for December-January-February (DJF) and June-July-August (JJA), overlaid with (contour) the 250 hPa zonal wind. The LWA in winter exhibits three maxima that reside over Europe, the northwest Atlantic Ocean, and the north and northwest Pacific Ocean. Interestingly, the European LWA maximum is dominated by northern anticyclonic wave activity akin to the blocking high in Figure 2a, while the northwest Atlantic LWA maximum represents southern cyclonic wave activity as exemplified by the cyclonic low in Figure 2b (not shown). In contrast, the LWA in summer is reduced in amplitude over the Atlantic and Pacific Oceans. Following the poleward shift of the North Atlantic jet from winter to summer, the European LWA maximum shifts to the north, joining a zonally elongated LWA band from the northwest Atlantic to Russia. Events of large wave amplitude over

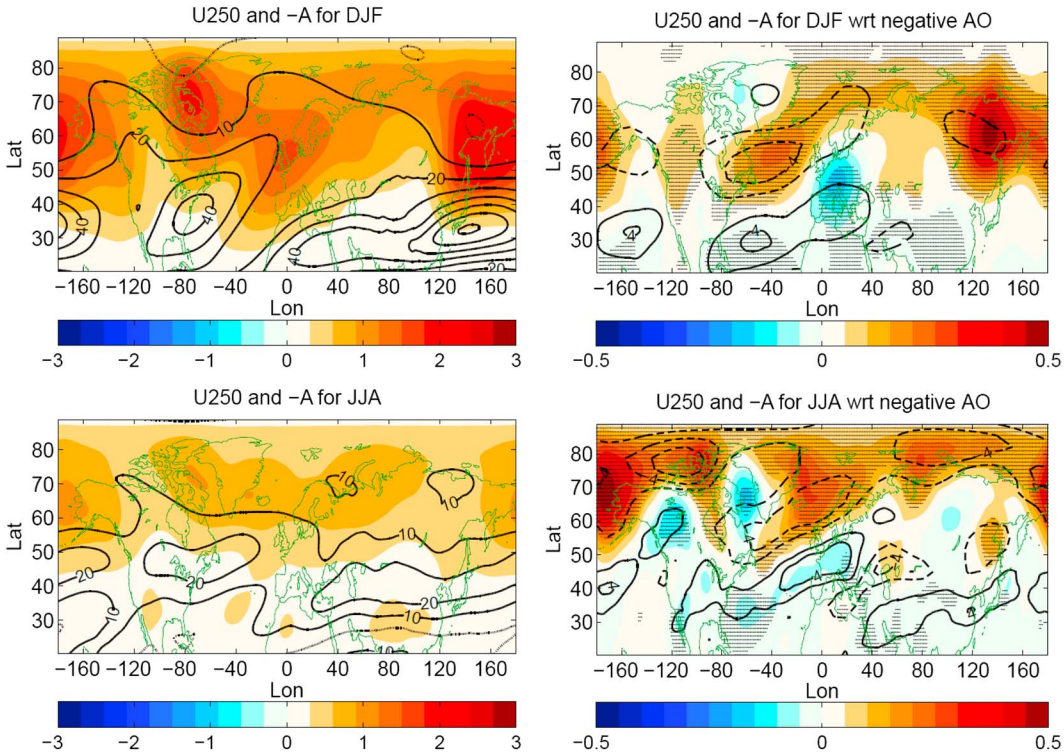


**Figure 3.** Zonal wind (250 hPa; contour) and inverted LWA of $Z_{500}$ (shading) for (top row) DJF and (bottom row) JJA: (left column) the 1979–2014 climatology and (right column) the regression pattern with respect to the inverted AO index, where hatching indicates the regions that are statistically significant at the 95% confidence level. Unit: m $s^{-1}$ for zonal wind and $10^8$ $m^2$ for LWA.

Europe and Russia are exemplified in Figures 2c and 2d. Overall, these LWA characteristics agree broadly with the winter and summer blocking frequency in *Masato et al.* [2013, Figures 1 and 5].

The regression pattern of LWA with respect to the AO index is shown in Figure 3 (right column). In winter, the negative phase of the AO is associated with a prominent meridional dipole in zonal wind over the North Atlantic. The negative zonal wind anomaly coincides with a large increase in wave activity across the Atlantic, with a positive zonal wind anomaly to the south and weakened wave activity over Spain, similar to the distribution of RWB frequency for the negative phase of the North Atlantic Oscillation (NAO) [*Woollings et al.*, 2008, Figure 2]. Another center of enhanced action is located over the northwest Pacific Ocean, corresponding to the northern mode of the East Asian winter monsoon (EAWM) and the associated Siberian cold air intrusions [e.g., *Wang et al.*, 2010]. In summer, the negative phase of the AO is characterized by a meridional dipole in zonal wind over the North Atlantic as in winter and a weakening of the zonal wind in high latitudes. Also, similar to winter, the negative zonal wind anomaly is correlated with largely increased wave activity, whereas the positive anomaly is correlated with weakened wave action. For a frictionless and adiabatic barotropic flow, the change in zonal mean zonal wind is anticorrelated with the change in wave activity amplitude (see equation (S11) in supporting information Text S1). *Huang and Nakamura* [2015] generalized this relationship as a function of longitude and found a local cancelation between the zonal wind and the LWA of PV in the upper troposphere during Hurricane Sandy. It is interesting to note that this anticorrelation also holds for the upper level zonal wind and the LWA of $Z_{500}$ associated with the AO.

One should note the differences between diagnoses of blocking and LWA. Blocking detection algorithms generally assign a threshold for the duration of an event, which is not considered for LWA, although we are currently investigating the duration of LWA as an additional criterion for identifying an extreme event. Blocking is often defined by the reversal of $Z_{500}$ [e.g., *Tibaldi and Molteni*, 1990], but the LWA accounts for both large and small displacements in $Z_{500}$. The central blocking position is often defined by the storm-track axis [*Masato et al.*, 2013], whereas the latitude of LWA is denoted by the equivalent latitude after the contour is adjusted to a zonally symmetric state. Because the definition of LWA (equation (6)) is merely a line integral and it shares

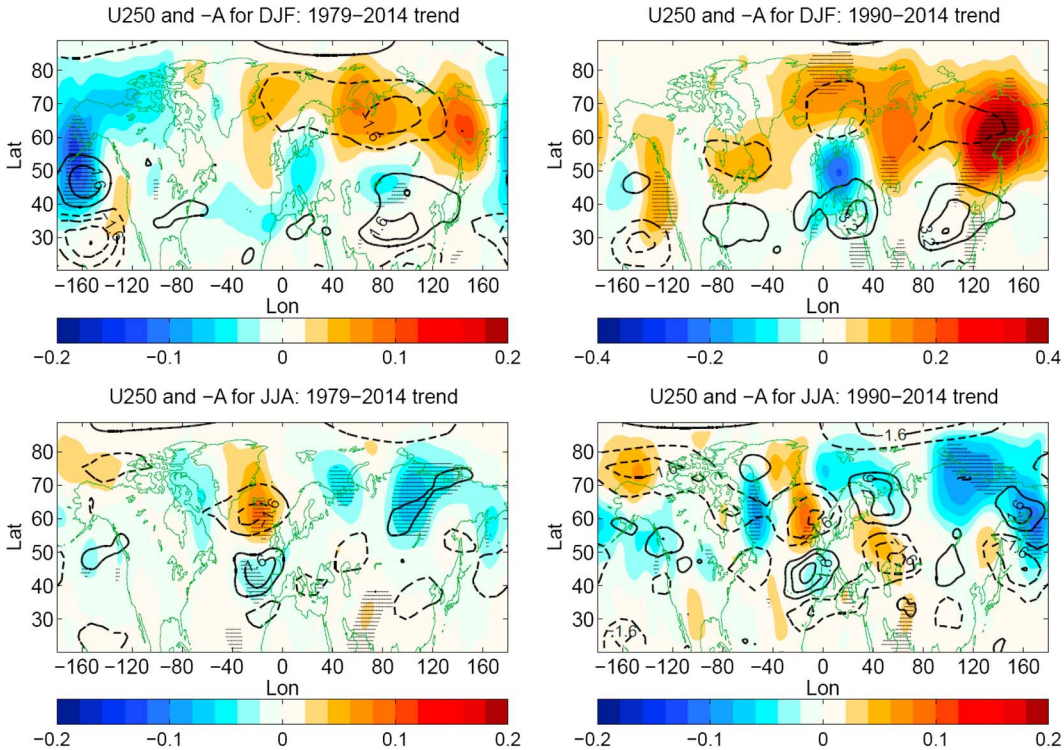


**Figure 4.** As in Figure 3 but for the linear trends in (left column) 1979–2014 and (right column) 1990–2014. Unit: m $s^{-1}$ $decade^{-1}$ for zonal wind and $10^8$ $m^2$ $decade^{-1}$ for LWA. Note that the color scale for the 1990–2014 trend in DJF is twice the scale of other panels.

many similarities with the blocking frequency reported in the literature, we assert that LWA is an objective diagnostic of midlatitude extreme weather, especially for those associated with atmospheric blocking.

We now switch to investigate the linear trends of LWA in recent decades. *Barnes et al.* [2014] explored the blocking trends in recent decades by applying three different blocking detection methods to four different reanalyses including ERA-I and NCEP2 examined here, and they found no clear hemispheric increases in blocking, although robust seasonal increases or decreases can be found over isolated regions. Since some of their uncertainties can result from subtle differences among detection methods [see *Barnes et al.*, 2014, Figure 1], we examine the linear trends in wave amplitude using our newly developed LWA and discuss the large-scale features in the context of previous studies on midlatitude extremes. Figure 4 presents the trends for the periods of 1979–2014 and 1990–2014, the latter representing the period with rapid Arctic warming and sea ice loss [e.g., *Francis and Vavrus*, 2012]. The regions where the trends are statistically significant at the 95% confidence level are indicated by hatching.

In winter, there is a general tendency of increased wave activity over Eurasia, but the trends are statistically significant only over Siberia for 1990–2014. This agrees with the robust increases in blocking over Asia from the two blocking detection methods of *Barnes et al.* [2014] and the Arctic sea ice influence on frequent Eurasian cold winters in recent decades in *Mori et al.* [2014]. There is also a significant increase in wave activity over the northeast Pacific Ocean for 1990–2014, which may be related to the positive trend in the atmospheric ridge there and the associated California drought [*Swain et al.*, 2014]. Nevertheless, the trends described above are not significant over a longer period of 1979–2014, which only indicates a decrease in wave activity over the Pacific Ocean.

In contrast, the summer trends in wave activity display two regions of significant change for both 1979–2014 and 1990–2014. First, there exists a meridional dipole in LWA over the North Atlantic, associated with a similar dipole in zonal wind. This supports a meridional shift in wave activity [*Barnes*, 2013] rather than an amplification in wave amplitude [*Francis and Vavrus*, 2012], although the northern region of the dipole with increased wave activity may be accompanied by more frequent extreme weather. Meanwhile, there is a weakening trend in wave activity over Russia that was also emphasized by *Coumou et al.* [2015]. In contrast to the conclusion

of *Coumou et al.* [2015], our diagnostic suggests less heat waves in that region due to reduced atmospheric waviness, but the reduction in wave amplitude may be offset by the rising global mean temperature, producing warmer temperature with less variability.

Finally, we briefly compare the results from ERA-I (Figures 3 and 4) with the same analysis using NCEP2 (supporting information Figures S1 and S2). Despite two different reanalysis products with different horizontal resolutions, not only the LWA climatology and its pattern associated with the AO agree remarkably well between the two reanalyses but the linear trends in LWA also display broadly consistent features for the individual periods examined. This again supports LWA as a robust diagnostic of midlatitude weather events.

## 5. Conclusions and Discussion

We have introduced the newly developed LWA of *Huang and Nakamura* [2015] as a diagnostic for characterizing midlatitude weather events. It is found that the LWA climatology and its variability associated with the AO are broadly consistent with the previously reported blocking frequency in the literature [*Pelly and Hoskins*, 2003; *Woollings et al.*, 2008; *Masato et al.*, 2013]. Compared with different blocking detection methods with adjustable parameters, LWA is solely based on the line integral of a dynamical quantity (equation (6)), and thus, it is established as an objective diagnostic of weather events.

Applying this quantification to the wave activity trends in the ERA-I and NCEP2 reanalyses, we found a strong seasonal and spatial dependence in the trends of wave amplitude in recent decades, as in *Barnes et al.* [2014]. We also found no statistically significant evidence either in winter or in summer, as seen in Figure 4, for a hemispheric-scale increase in wave amplitude associated with recent Arctic warming that was hypothesized by *Francis and Vavrus* [2012]. Interestingly, we found a local anticorrelation between the zonal wind anomaly and LWA amplitude for the patterns associated with the AO as well as for the linear trend. This is expected for the zonal mean fields associated with the eddy-zonal flow interaction in a barotropic flow (see equation (S11) in supporting information Text S1), but the baroclinic process is likely to play an important role for the long-term trend due to the polar amplification of global warming.

LWA can identify robust changes in wave amplitude at the regional scales that can be discussed in the context of previous studies on midlatitude extremes. In winter, increased wave activity is identified by LWA over Siberia [*Mori et al.*, 2014] and over the northeast Pacific Ocean [*Swain et al.*, 2014] for the period of 1990–2014, while a decrease in wave activity is found over the Pacific Ocean for 1979–2014, likely as the result of the large decadal variability there. In contrast, the summertime trends in wave activity display a dipole structure over the North Atlantic [*Barnes*, 2013], and weakened eddy action in Russia [*Coumou et al.*, 2015] for both periods of 1979–2014 and 1990–2014.

Given the lucid linkage to weather extremes (Figure 2), further investigations are warranted into the application of LWA to understanding midlatitude extremes in a changing climate. The LWA theory, with the generalized EP flux [*Huang and Nakamura*, 2015], provides a new framework into the mechanisms of the onset, duration, and breakdown of atmospheric blocking, heat waves, or atmospheric rivers. Meanwhile, LWA can be used as an objective metric to identifying the forced trends in wave amplitude in climate models in response to climate change.

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